Utilizing Sales Pattern Analytics to Customize Marketing and Optimize Stock at a Pharmacy

Final Report for the BDM Capstone Project

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1. Executive Summary

[Pharmacy] is a small-scale retail pharmacy in Bengaluru , selling prescription medicines, over-the-counter products, health supplements, and personal-care items. Before this project, the store had no systematic way of knowing which customers to retain, which products to protect against stock-outs, or which items to promote together. Decisions were made on instinct, not evidence. This report presents the completed final-phase analysis addressing that gap: customer segmentation, product association analysis, demand forecasting, and inventory prioritisation, applied to six months of the store's own transaction data (1 January–30 June 2024; 25,512 line records; ₹9,195,287.54 total revenue).

The central finding is that both revenue and customer relationships are concentrated. A segment of 213 customers (25.98% of the 820-customer base) generates 40.85% of revenue and, together with a second Steady/Regular segment, accounts for 77.70% of revenue from 63.66% of customers, leaving a clear, addressable group of at-risk customers whose lapse would be costly. On the product side, 15 of 40 products drive 69.85% of revenue, and of these, five (Salbutamol Inhaler, Vitamin D3 60K, Volini Spray, Calcium + D3, and Montelukast) combine high revenue with unstable month-to-month demand, making them the store's single biggest stock-out risk; this report proposes specific safety-stock and reorder-point figures for each. Product-association analysis found that chronic-therapy medicines are frequently bought together, with the eight strongest pairings collectively touching 22.29% of total revenue, a concrete cross-selling and bundling opportunity.

A trend-based forecast of near-term revenue was validated against actual results by training the model on five months and testing it against the sixth: the resulting error was 3.73% (MAPE) at the overall revenue level, giving reasonable confidence in short-term financial planning, though category-level forecasts vary widely in reliability and should be read with that caveat. Together, these findings give [Pharmacy] a specific, numbers-backed basis for customer retention, inventory safety stock, and cross-sell promotions, directly resolving the three problem statements set out in the original project proposal.

2. Detailed Explanation of the Analysis Process / Method

2.1 Data Collection

The primary quantitative data for this study was obtained directly from [Pharmacy] , with written authorisation from the proprietor permitting its use strictly for academic purposes as part of the BDM capstone project. The transaction workbook supplied by the pharmacy contains a transaction-level sheet of 25,512 rows and 18 columns, and a summary sheet, together covering invoice, customer, product, category, brand, batch, expiry, quantity, pricing, tax, revenue, and payment-mode fields for the six-month period from 1 January 2024 to 30 June 2024.

Supporting evidence for the final document includes the data sheet with analyses and relevant charts which can be found here:

[link removed]

2.2 Data Understanding

The transaction dataset comprises 25,512 transaction-line records spanning 13,355 unique invoices, 820 unique customers, 40 unique products, 12 brands, and 14 therapeutic categories. Table 2.1 sets out the data types, and Table 2.2 links the key variables to the three problem

statements defined in the proposal, so that each analysis in this report can be traced back to a specific business objective.

Table 2.1 Data Types in the Transaction Dataset

Data Type	Examples	Unit / Format
Categorical	Category, Brand, Product_Name, Payment_Mode, Customer_ID, Product_ID	Text labels
Numerical	Quantity, Unit_Price, Discount, GST, Total_Amount	Units, ₹, ₹, %, ₹
Date & Time	Invoice_Date, Invoice_Time, Expiry_Date	DD-MM-YYYY, HH:MM
Identifier / Text	Invoice_No, Customer_Name, Batch_No	Text / alphanumeric code

Table 2.2 Mapping of Dataset Variables to Problem Statements

Problem Statement	Key Variables Used	Analysis Applied
PS1: Purchase trends & repeat-buying behaviour	Customer_ID, Invoice_No, Invoice_Date, Total_Amount	RFM segmentation, descriptive frequency analysis
PS2: Customer categorisation for targeted marketing	Customer_ID, Total_Amount, Invoice_Date, Payment_Mode	RFM segmentation, segment-wise revenue analysis
PS3: Inventory planning & cross-selling	Product_ID, Category, Quantity, Total_Amount, Invoice_No	Market basket analysis, ABC / ABC-XYZ classification, demand forecasting

Dataset profiling confirmed that invoice numbers group multiple product lines into complete customer transactions, enabling both invoice-level basket analysis and customer-level aggregation for RFM segmentation. Date fields support daily and monthly demand comparisons required for the forecasting analysis.

2.3 Data Preprocessing

The transaction workbook was checked for missing values, duplicate rows, data types, and invoice-level consistency. All 25,512 rows contained values across all 18 columns, and no exact duplicate rows were found. Invoice dates were converted to date-time values and grouped into calendar months and days. For the final-phase analyses, invoice-level product baskets were reconstructed from invoice numbers, customer-level aggregates (recency, frequency, monetary value) were computed per Customer_ID, product-level revenue and monthly quantity were aggregated per Product_Name, and daily demand series were built for products selected for safety-stock calculation.

2.4 Descriptive Statistical Analysis

Descriptive statistics were calculated on invoice value (Total_Amount summed per invoice) and basket size (number of product lines per invoice) to characterise the scale and spread of individual transactions, extending beyond the mean/median/standard-deviation view reported at the midterm stage.

Table 2.3 Descriptive Statistics of Invoice Value (₹)

Statistic	Value
Count (invoices)	13,355
Mean	₹688.53
Median	₹395.08
Standard Deviation	₹929.99
Minimum	₹14.28
Maximum	₹10,466.40
Range	₹10,452.12
1st Quartile (Q1)	₹190.40
3rd Quartile (Q3)	₹836.84
Coefficient of Variation	135.07%

The mean invoice value (₹688.53) sits well above the median (₹395.08), and the coefficient of variation of 135.07% indicates very high relative spread: most invoices are small, low-value purchases (Q1 = ₹190.40), while a smaller number of large, high-value invoices (up to ₹10,466.40) pull the average upward. This right-skewed pattern is consistent with a pharmacy that serves frequent, low-cost repeat purchases (chronic medicine refills) alongside occasional large, multi-item transactions, and it is the reason RFM segmentation, which separates customers by their typical spend instead of relying on a single average, is the appropriate tool for customer analysis in Section 2.5.

Basket size averaged 1.91 items per invoice (median 2, mode 1, maximum 4), confirming that most invoices are small but a meaningful share involve multiple products, which is the base population for the market basket analysis in Section 2.6.

2.5 Customer Segmentation (RFM) Methodology

Recency, Frequency, and Monetary (RFM) analysis was used to segment the 820 customers. Recency was calculated as the number of days between each customer's most recent invoice and one day after the last date in the dataset (30 June 2024); Frequency was the count of unique invoices per customer; Monetary was each customer's total spend. Each measure was scored into quartiles (1–4, with 4 indicating the most favourable value), and the three scores were summed to produce a combined RFM score used to assign customers to one of four segments: High-Value Loyal, Steady/Regular, Occasional, and At-Risk/Infrequent. This directly addresses Problem Statement 1 (purchase trends and repeat-buying behaviour) and Problem Statement 2 (customer categorisation for targeted marketing) by converting raw invoice history into defined, actionable groups instead of a single aggregate repeat-purchase figure.

2.6 Market Basket Analysis Methodology

Market basket analysis was performed on the 7,399 invoices containing two or more product lines. Invoice-level product baskets were encoded into a one-hot transaction matrix, and the Apriori algorithm was applied with a minimum support threshold of 1% to identify frequently co-purchased product sets. Association rules were then generated and ranked by support, confidence, and lift, which together identify specific product pairs that are purchased together more often than chance would predict. This addresses Problem Statement 3's cross-selling objective with named product combinations, going beyond the general observation that multi-product invoices exist.

2.7 Demand Forecasting Methodology

A linear trend regression was chosen to project near-term demand, fitted separately to monthly total revenue and to monthly unit demand for each therapeutic category. This method was selected over more complex seasonal models because six months of history is too short to reliably estimate a seasonal component; a transparent linear trend avoids overfitting to a small number of observations.

To evaluate forecast accuracy instead of simply asserting it, the model was validated using a holdout approach: it was trained on the first five months (January–May) and used to predict the sixth month (June), which was then compared against the actual observed June value using Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). This validation is reported alongside the forecast in Section 3.4 so that the reliability of each projection is stated explicitly instead of assumed.

2.8 ABC and ABC-XYZ Inventory Classification Methodology

ABC analysis ranked all 40 products by six-month revenue in descending order and calculated cumulative revenue contribution, classifying products as Class A (cumulative revenue up to 70%), Class B (70–90%), and Class C (the remaining 90–100%). Revenue-based ABC classification alone does not capture how predictable a product's demand is, so it was combined with an XYZ classification: the coefficient of variation (CV) of each product's monthly unit demand was calculated, and products were classified as X ($CV \leq 20\%$, stable demand), Y ($20\% < CV \leq 40\%$, moderately variable demand), or Z ($CV > 40\%$, highly variable demand). Combining the two (e.g. AY = high-revenue but variable-demand) identifies the specific products where a high revenue share is compounded by unpredictable demand: the products most likely to cause a costly stock-out, a distinction that plain ABC analysis alone would not surface.

2.9 Tools Used and Data Visualisation

Microsoft Excel was used as the supplied transaction-data source and for initial validation. Python (Pandas, NumPy, mlxtend for Apriori-based market basket analysis, and Matplotlib) was used for data cleaning, RFM computation, association-rule mining, trend-based forecasting with holdout validation, ABC/XYZ classification, and chart generation. The visualisations in this report were generated directly from the transaction workbook.

3. Results and Findings

This section presents the completed final-phase analyses (customer segmentation, market basket analysis, demand forecasting, and ABC/ABC-XYZ inventory classification), each

linked to a measurable business outcome, building on a brief recap of the descriptive baseline first reported at the midterm stage.

3.1 Descriptive Sales Overview

As established at the midterm stage, total revenue over the six-month period was ₹9,195,287.54, with Diabetes (30.65%), Vitamins (11.08%), Cardiac (10.71%), and Respiratory (10.14%) as the leading revenue categories. Monthly revenue was broadly stable, ranging from ₹14,69,148.80 (April) to ₹16,11,921.74 (March). These midterm findings, based on invoice counts and category totals, are descriptive only; the customer- and product-level groupings that make them actionable are developed in Sections 3.2–3.5 below.

3.2 Customer Segmentation Results (RFM)

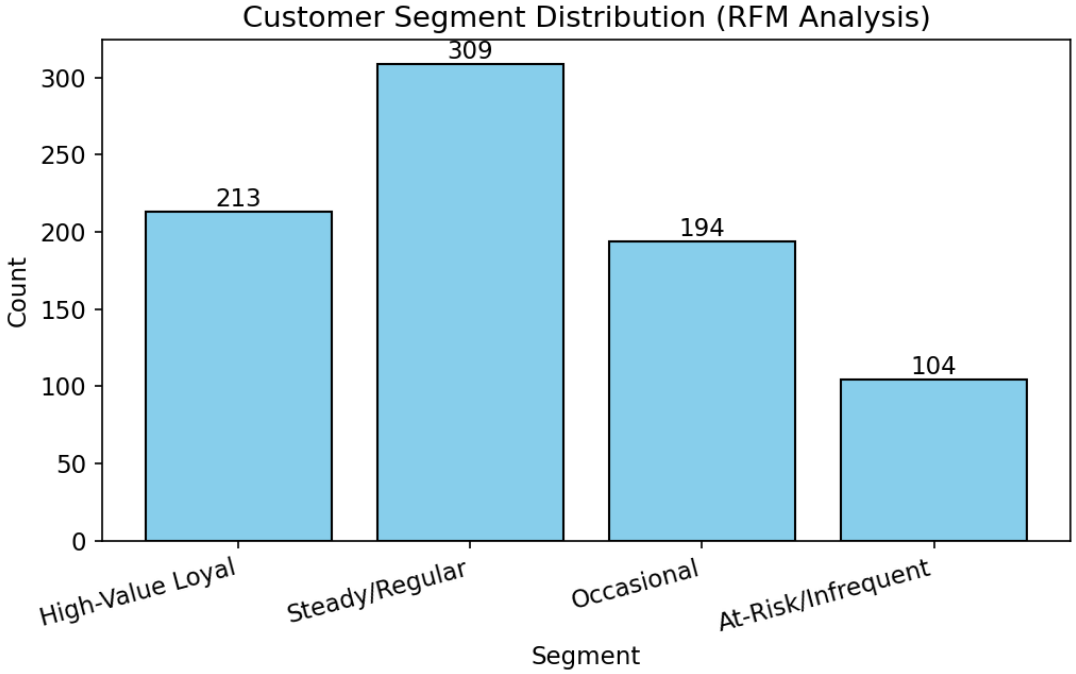


Figure 1: Distribution of customers across RFM-derived segments.

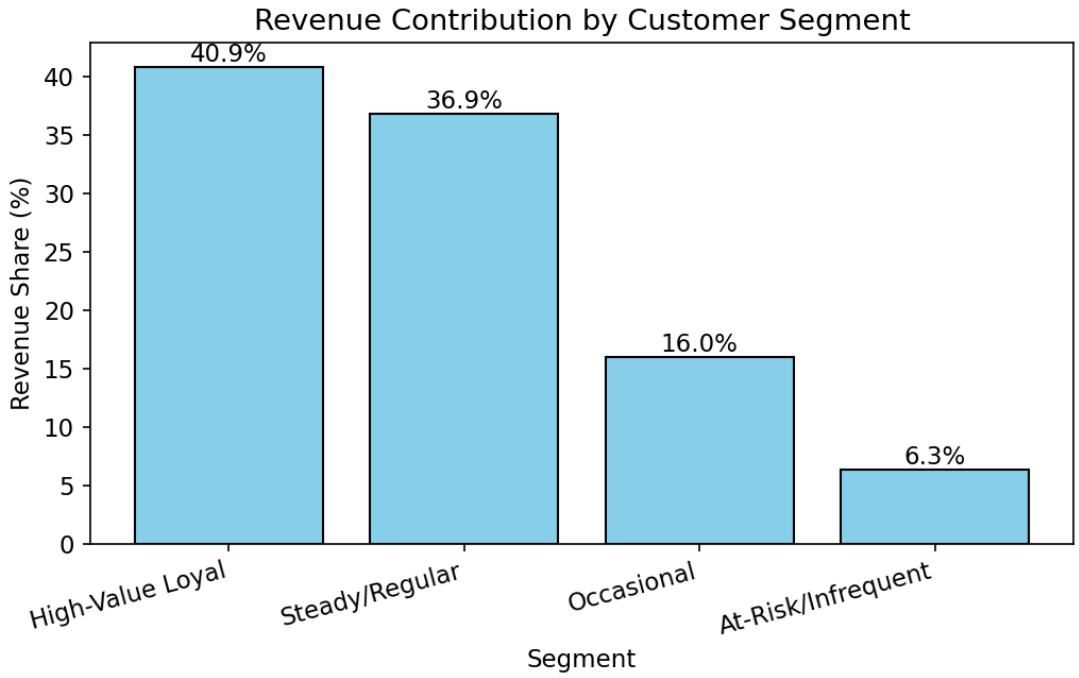


Figure 2: Revenue contribution by customer segment.

Table 3.1 RFM Segment Summary

Segment	Customers	% of Customers	Avg. Recency (days)	Avg. Frequency (invoices)	Avg. Monetary (₹)	Revenue Share
High-Value Loyal	213	25.98%	6.08	21.59	17,636.98	40.85%
Steady/Regular	309	37.68%	9.90	16.49	10,965.65	36.85%
Occasional	194	23.66%	14.96	12.99	7,577.18	15.99%
At-Risk/Infrequent	104	12.68%	25.00	10.98	5,579.34	6.31%

Observation

The High-Value Loyal segment (213 customers, 25.98% of the customer base) purchases most recently (average 6.08 days since last visit), most frequently (21.59 invoices on average), and spends the most (₹17,636.98 on average), together generating 40.85% of total revenue. The Steady/Regular segment (309 customers, the largest group) contributes a further 36.85% of revenue. Together, these two segments (63.66% of customers) account for 77.70% of total revenue. The At-Risk/Infrequent segment (104 customers, 12.68%) has not purchased in an average of 25 days and contributes only 6.31% of revenue. This segmented breakdown replaces the single midterm-stage statistic that 99.51% of customers had more than six invoices. That figure, on its own, does not distinguish a loyal customer from an infrequent one; the RFM segments make that distinction explicit.

Interpretation

The concentration of revenue in a relatively small, highly engaged customer base confirms the repeat-purchase pattern identified at the midterm stage and gives it a segmented, actionable structure. The clear separation between the High-Value Loyal and At-Risk/Infrequent segments indicates that customers follow different re-purchase cycles that a uniform, one-size-fits-all marketing approach would not address.

Business Outcome

This analysis directly supports a measurable retention target: protecting the 40.85% of revenue held by the High-Value Loyal segment, and converting a portion of the 104 At-Risk customers (currently 6.31% of revenue) back into the Steady/Regular tier through win-back offers, which Section 4.3 quantifies further.

3.3 Market Basket Analysis Results

Of the 13,355 invoices in the dataset, 7,399 (55.40%) contain two or more product lines, providing a substantial base for association-rule mining. Applying the Apriori algorithm with a 1% minimum support threshold to these multi-product invoices identified 76 frequent itemsets, from which association rules were generated and ranked by lift.

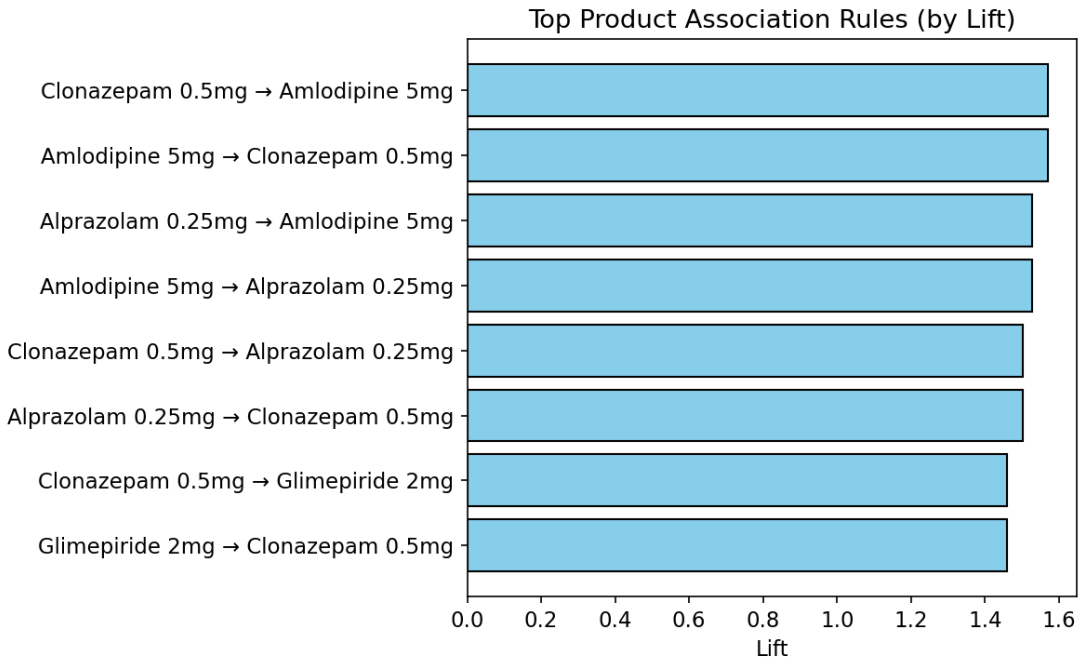


Figure 3: Top product association rules ranked by lift.

Table 3.2 Top Product Association Rules

Product A	Product B	Support	Confidence	Lift
Clonazepam 0.5mg	Amlodipine 5mg	2.01%	17.63%	1.57
Alprazolam 0.25mg	Amlodipine 5mg	1.95%	17.16%	1.53
Clonazepam 0.5mg	Alprazolam 0.25mg	1.95%	17.04%	1.50
Clonazepam 0.5mg	Glimepiride 2mg	1.89%	16.57%	1.46
Atorvastatin 10mg	Metoprolol 50mg	1.81%	15.95%	1.45

Product A	Product B	Support	Confidence	Lift
Levothyroxine 50mcg	Atorvastatin 10mg	1.70%	16.26%	1.43
Glimepiride 2mg	Metformin 500mg	1.82%	16.07%	1.43
Atorvastatin 10mg	Glimepiride 2mg	1.82%	16.07%	1.42

Observation

The strongest associations (lift 1.42–1.57) occur among chronic-therapy medicines: sleep/anxiety medicines (Clonazepam, Alprazolam) are frequently co-purchased with cardiac and diabetes medicines (Amlodipine, Glimepiride), and cardiac medicines (Atorvastatin, Metoprolol) are co-purchased with each other and with thyroid and diabetes medicines (Levothyroxine, Metformin, Glimepiride). All reported pairs have lift greater than 1, meaning these products are purchased together more often than would be expected by chance. Collectively, the eight products involved in these top rules generate ₹20,49,455.80 in revenue, 22.29% of total revenue, marking a concrete, sizeable base for cross-selling activity, beyond the general observation that 55.40% of invoices contain multiple products.

Interpretation

The associations suggest that a meaningful share of [Pharmacy] 's customers manage multiple chronic conditions simultaneously (for example, cardiac disease alongside diabetes, or anxiety/sleep disorders alongside hypertension), and refill these combinations together on a recurring basis. This is consistent with the highly repeat-oriented High-Value Loyal and Steady/Regular segments identified in Section 3.2.

Business Outcome

[Pharmacy] can use these eight named pairs to design combination refill reminders and adjacent counter placement, with a realistic near-term goal of lifting average basket size (currently 1.91 items per invoice) toward 2.2–2.3 items by prompting the paired purchase at billing time, which would add a measurable increment to average invoice value (currently ₹688.53).

3.4 Demand Forecasting Results

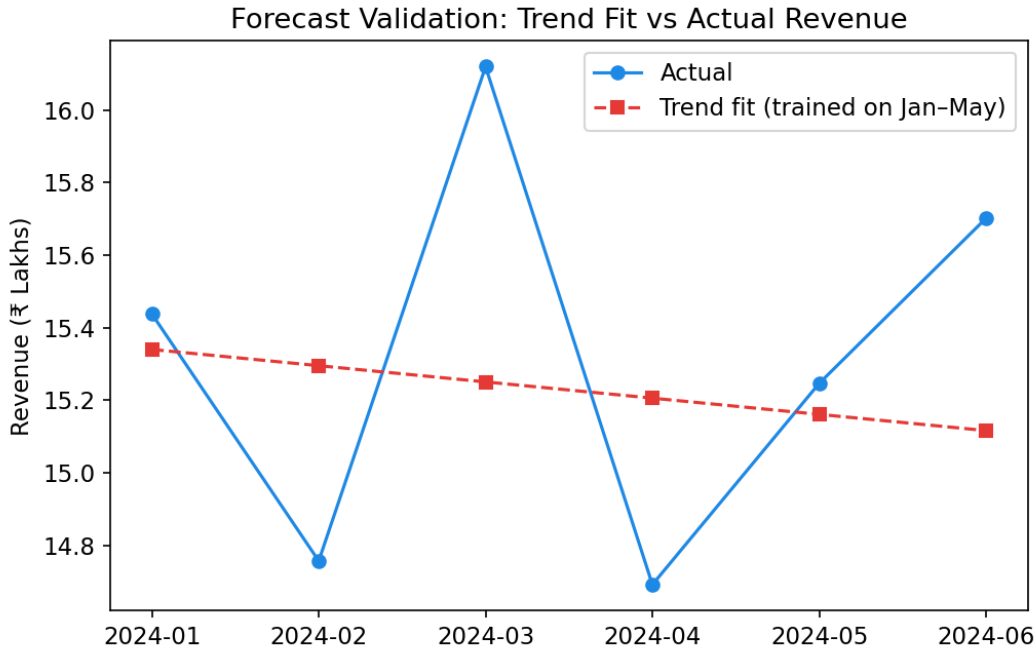


Figure 4: Forecast validation: trend fitted on January–May compared against actual June revenue.

The linear trend model, trained on January–May and tested against the actual June figure, predicted ₹15,11,608.92 against an actual ₹15,70,149.28, a Mean Absolute Error of ₹58,540.36 and a Mean Absolute Percentage Error (MAPE) of 3.73%. This level of error is small relative to the month-to-month range observed across the six months, giving reasonable confidence in the model for short-term overall revenue planning.

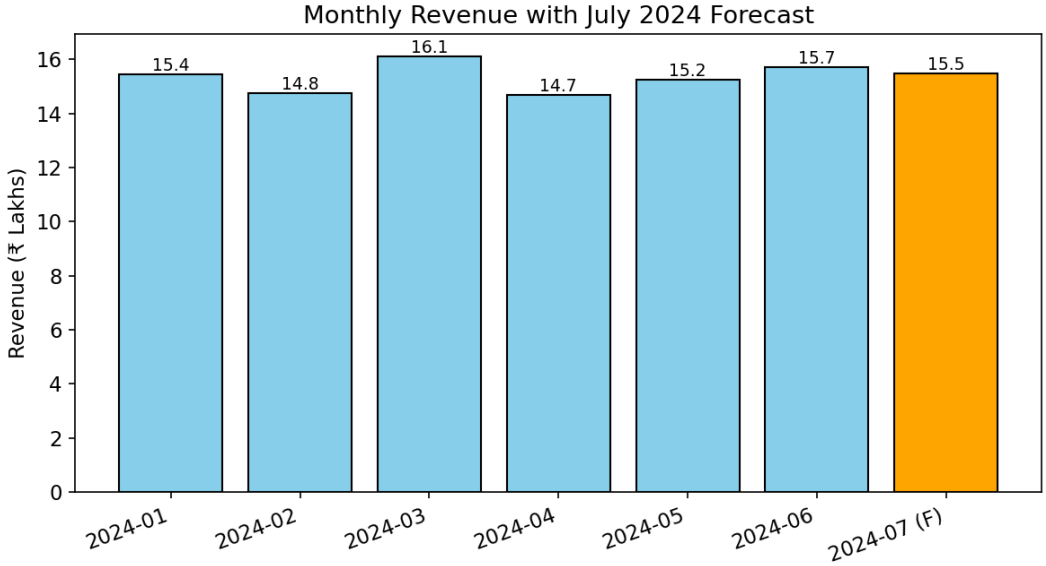


Figure 5: Monthly revenue, January–June 2024, with the full-data linear-trend forecast for July 2024.

Using all six months, the trend projects July 2024 revenue of approximately ₹15.46 Lakhs, consistent with the stable, narrow-range pattern observed throughout the period.

Table 3.3 Category-Level Forecast Validation (Holdout MAPE, June 2024)

Category	Holdout MAPE	Forecast Reliability
Diabetes	2.4%	High
Eye/Ear	6.3%	High
Neuro/Sleep	8.4%	High
OTC	16.7%	Moderate
Cardiac	18.1%	Moderate
Skin Care	59.9%	Low
Respiratory	60.8%	Low
Vitamins	62.0%	Low

Observation

Forecast reliability differs sharply by category. Diabetes, Eye/Ear, and Neuro/Sleep demand can be predicted with holdout error under 10%, meaning the linear trend is a reasonably dependable planning input for these categories. Vitamins, Respiratory, and Skin Care, by contrast, show holdout errors above 55%, meaning the six-month trend is not yet reliable for these categories and any procurement decision based on it should be treated as indicative only, not confirmed. This directly reflects the demand-variability finding in the ABC-XYZ analysis in Section 3.5: the categories with poor forecast accuracy are the same ones populated by Y-class (variable-demand) products.

Interpretation

Overall revenue is forecast to remain within the same stable band observed over the six-month history, and this projection can be trusted to within about 4%. Category-level projections cannot be extended with the same confidence; six months is too short a history to separate a genuine trend from month-to-month noise for the more volatile categories, and this report treats those figures as preliminary monthly patterns, not confirmed seasonal trends.

Business Outcome

Procurement decisions for Diabetes, Eye/Ear, and Neuro/Sleep products can reasonably use the trend forecast directly. For Vitamins, Respiratory, and Skin Care, [Pharmacy] should continue using the ABC-XYZ-based safety-stock approach in Section 3.5 instead of the trend forecast alone, and should revisit the seasonal pattern once a full 12-month history is available.

3.5 ABC and ABC-XYZ Analysis Results

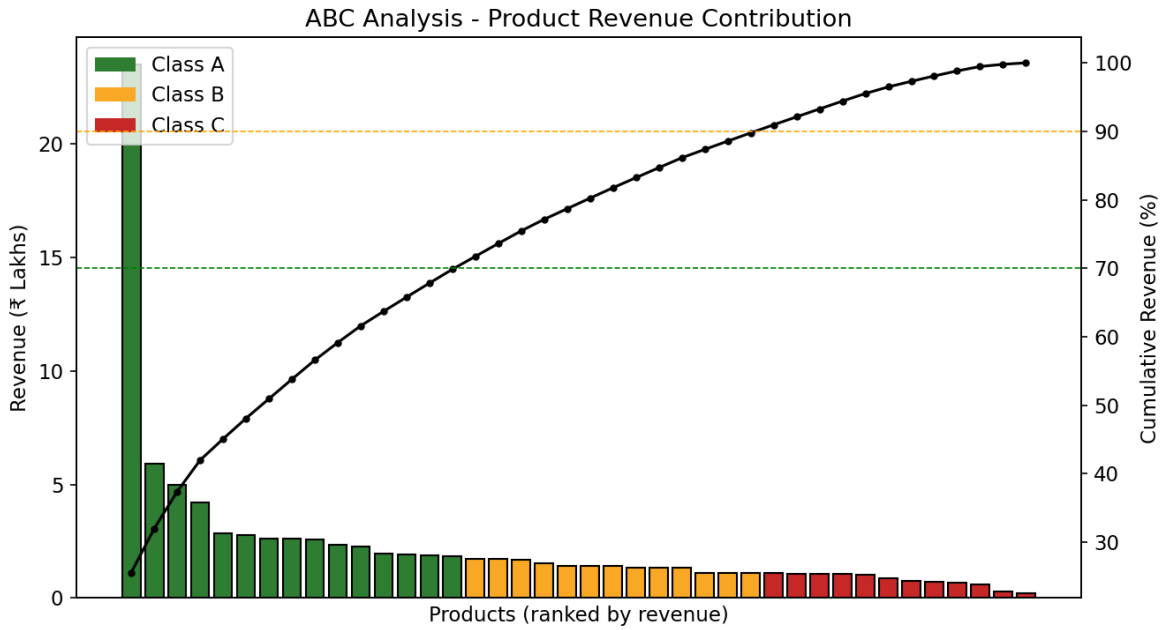


Figure 6: ABC Pareto chart: products ranked by revenue with cumulative revenue contribution.

Table 3.4 ABC Classification Summary

Class	Products	% of Products	Revenue (₹)	% of Revenue
A	15	37.5%	64,23,306.54	69.85%
B	13	32.5%	18,31,062.33	19.91%
C	12	30.0%	9,40,918.67	10.23%

Insulin Glargine alone contributes 25.57% of total revenue, and the 15 Class-A products together account for 69.85% of revenue from just 37.5% of the product range, refining the midterm finding that the top eight products contribute 53.79% of revenue by extending the classification to the full 40-product range.

ABC-XYZ Combined Classification

Revenue share alone does not indicate how predictable a product's demand is. Combining ABC revenue class with an XYZ demand-variability class (X = stable, CV ≤ 20%; Y = moderately variable, 20–40%; no product exceeded the Z threshold of 40% CV) produces a more complete inventory picture.

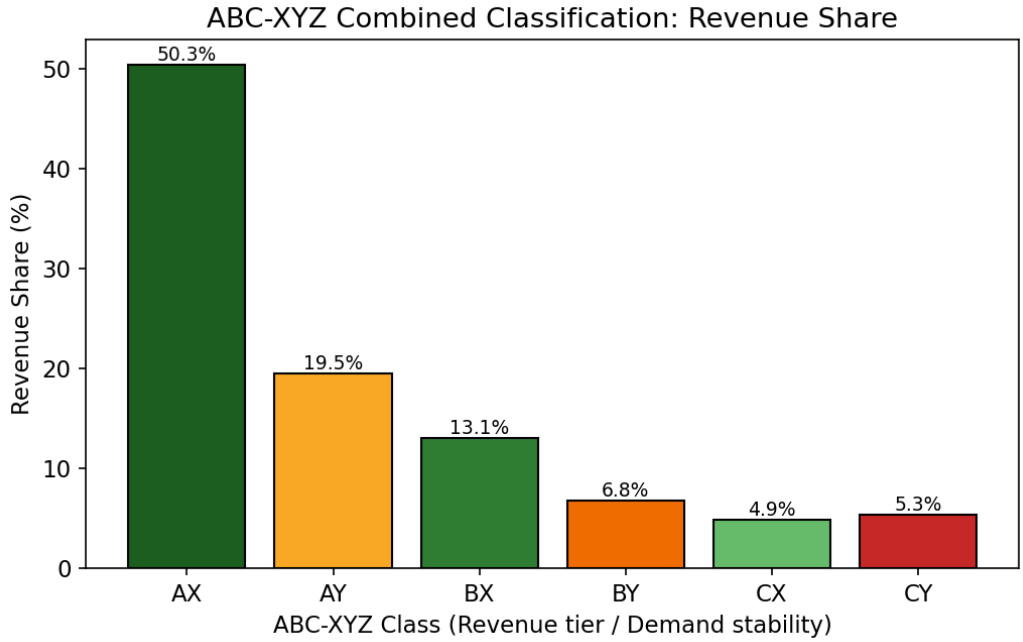


Figure 7: Revenue share by combined ABC-XYZ class.

Table 3.5 Class-A Products with Unstable Demand (AY): Priority for Safety Stock

Product	Revenue (₹)	Demand CV	Avg. Daily Demand (units)	Safety Stock (units)	Reorder Point (units)
Salbutamol Inhaler	5,90,009.28	32.65%	7.9	19	43
Vitamin D3 60K	4,21,835.84	21.35%	6.4	15	34
Volini Spray 100g	2,84,619.99	23.48%	7.4	19	41
Montelukast 10mg	2,35,000.92	26.97%	7.2	16	38
Calcium + D3	2,62,144.08	21.42%	7.0	16	38

Observation

10 products fall into AX (high revenue, stable demand; 50.35% of total revenue) and can be managed with lean, predictable stock levels. Five products fall into AY (high revenue, variable demand; 19.51% of total revenue): Salbutamol Inhaler, Vitamin D3 60K, Volini Spray, Montelukast, and Calcium + D3. These five combine revenue importance with demand unpredictability and are therefore the store's highest stock-out risk. Using each product's daily demand standard deviation, a 3-day replenishment lead time, and a 95% service level (Z = 1.65), the safety stock and reorder point in Table 3.5 give specific, calculable stocking targets instead of a general instruction to "monitor closely." No product in the dataset fell into the Z (highly variable) tier.

Interpretation

The AX/AY split shows that [Pharmacy] 's inventory risk is not spread evenly across its 15 Class-A products: half can be run lean, while the other third (5 of 15) need an explicit safety-

stock buffer. This distinction would be invisible from ABC revenue ranking alone, which is why the ABC-XYZ combination was added in this final phase.

Business Outcome

Applying the reorder points in Table 3.5 (34–43 units depending on product) to the five AY products gives [Pharmacy] a concrete stock-out-prevention target: maintaining these reorder levels should materially reduce the risk of running out of the store's highest-risk, high-revenue products, without the cost of carrying the same safety margin across all 40 products.

3.6 Data Quality Findings

Observation

Consistent with the midterm assessment, the transaction workbook contained no missing values across its 25,512 rows and 18 columns, and no exact duplicate rows. Product, category, and payment-mode fields were already standardised in the supplied data, and invoice numbers correctly grouped multi-product transactions, so no additional cleaning was required beyond date-type conversion and the aggregation steps described in Section 2.3.

Interpretation

The consistently clean structure of the transaction data meant that analytical effort in the final phase could be directed fully toward segmentation, association mining, forecasting, and classification instead of remedial data cleaning, and it supports confidence in the reliability of the reported findings.

Business Outcome

[Pharmacy] 's existing billing process already captures transaction data in a consistent, analysis-ready format; maintaining this discipline as transaction volume grows, and extending the history to a full 12 months, will allow the category-level forecasts in Section 3.4 to be re-validated with genuine seasonal data.

3.7 Summary of Findings

The final-phase analysis of [Pharmacy] 's transaction data completes the three problem statements set out in the proposal. Customer segmentation shows that 213 High-Value Loyal customers (25.98%) drive 40.85% of revenue, while 104 customers (12.68%) are at risk of lapsing. Market basket analysis identified eight specific, high-lift product pairs among chronic-therapy medicines, together touching 22.29% of revenue. Trend-based forecasting, validated by holdout testing, gives a reliable overall-revenue projection (3.73% MAPE) but only moderate-to-low reliability at the category level for Vitamins, Respiratory, and Skin Care. ABC-XYZ analysis identifies five specific products (AY class) that combine high revenue with unstable demand, for which specific safety-stock and reorder-point figures are proposed.

4. Interpretation of Results and Recommendations

4.1 Interpretation of Results

Taken together, the final-phase analyses complete the three problem statements defined in the original project proposal. The RFM segmentation directly addresses Problem Statement 1 (purchase trends and repeat-buying behaviour) and Problem Statement 2 (customer categorisation for targeted marketing) by translating the customer base's repeat-purchase

pattern, first visible at the midterm stage, into four segments with distinct engagement levels and revenue contributions. This goes beyond the single, less-informative statistic that most customers have more than six invoices.

The market basket and ABC-XYZ analyses address Problem Statement 3 (inventory planning and cross-selling) with specific, named product pairs and specific reorder-point figures, going beyond the midterm-stage observation that a majority of invoices contain multiple products. The demand forecast adds a forward-looking dimension, but this report is explicit that its reliability is uneven: the six-month history supports confident planning for Diabetes, Eye/Ear, and Neuro/Sleep, and only preliminary, unconfirmed patterns for Vitamins, Respiratory, and Skin Care, where the same six months are too short to distinguish a real seasonal trend from ordinary month-to-month variation.

4.2 Overall Interpretation

The findings collectively show that [Pharmacy] 's business, while operationally simple, already generates transaction data rich enough to support systematic, evidence-based decision-making. Revenue and customer value are both concentrated: in a defined set of high-priority products, and in a defined set of high-value, repeat customers. This means that a relatively small number of targeted actions (loyalty communication for two customer segments, safety stock for five specific products, and bundled promotions for eight specific medicine pairs) can address a large share of the business's marketing and inventory needs. Equally, the analysis is explicit about its limits: category-level seasonality and the AX-only products should be re-checked once a longer transaction history is available.

4.3 Recommendations

4.3.1 Prioritise Retention for High-Value and Steady Segments

[Pharmacy] should introduce simple loyalty recognition (for example, priority billing or small refill discounts) for the High-Value Loyal and Steady/Regular segments, which together represent 63.66% of customers and 77.70% of revenue, to protect this core revenue base.

4.3.2 Launch Win-Back Communication for At-Risk Customers

The At-Risk/Infrequent segment (104 customers, average 25 days since last purchase, currently 6.31% of revenue) should be targeted with refill reminders or reactivation offers via SMS/WhatsApp before these customers lapse further; converting even a quarter of this segment into the Occasional or Steady tier would be a measurable, trackable retention outcome for the next reporting period.

4.3.3 Bundle the Eight Named Co-Purchased Medicine Pairs

The eight specific association-rule pairs identified in Section 3.3 (for example, Clonazepam with Amlodipine, and Atorvastatin with Metoprolol) should inform combination refill reminders and adjacent counter placement, with a target of lifting average basket size from 1.91 toward 2.2–2.3 items per invoice.

4.3.4 Apply the Proposed Safety Stock and Reorder Points to the Five AY Products

Salbutamol Inhaler, Vitamin D3 60K, Volini Spray, Montelukast, and Calcium + D3 should be reordered at the specific stock levels in Table 3.5 (34–43 units, depending on product), instead of receiving the generic “monitor closely” treatment applied to all Class-A products; the

remaining ten AX products can be run at leaner, standard stock levels given their demand stability.

4.3.5 Use the Validated Forecast Only Where It Is Reliable

Procurement for Diabetes, Eye/Ear, and Neuro/Sleep can use the trend forecast directly, given holdout error under 10%. For Vitamins, Respiratory, and Skin Care, where holdout error exceeded 55%, [Pharmacy] should rely on the ABC-XYZ safety-stock approach in place of the trend line, and should revisit these forecasts once 12 months of data are available to test for genuine seasonality.

4.3.6 Develop a Management Dashboard

The RFM segments, association rules, ABC-XYZ classes, and validated forecasts developed in this project provide sufficient structure to support a lightweight dashboard (for example, in Power BI or Excel) that management can refresh monthly as new transaction data becomes available, including re-running the holdout validation each month to track whether forecast accuracy is improving as more history accumulates.

4.4 Conclusion

This final report distinguishes clearly between what was established at the midterm stage and what has been completed in this final phase. The midterm stage produced descriptive findings only: monthly and category revenue totals, top-product rankings by revenue, and the invoice-count-based observation that most customers had more than six invoices. Those findings, while directionally useful, did not yet segment customers, identify specific product associations, validate any forecast, or combine revenue with demand variability for inventory planning. These were the gaps identified in the midterm review.

This final phase closes each of those gaps with a specific, evidence-backed result: RFM segmentation into four named customer groups with quantified revenue shares; market basket analysis producing eight specific, lift-ranked product pairs; a demand forecast whose accuracy was tested by holdout validation (3.73% MAPE overall, with category-level accuracy reported honestly rather than assumed); and an ABC-XYZ classification that identifies five specific products needing explicit safety stock, with calculated reorder points. In practical terms, this final phase turns [Pharmacy] 's own transaction history into a working decision tool: specific customers worth retaining, specific products worth stocking more carefully, and specific combinations worth promoting together, closing out the three problem statements defined in the original project proposal.